

Alibaba takes open source route in the face of US-China trade war, launching RISC-V-based processor

Open source technologies are becoming an increasingly attractive option to Chinese companies amid the ongoing US-China trade war.

Chinese e-commerce giant Alibaba will no longer need to license any core from ARM or other foreign companies because it has now designed its own core based on the RISC-V. (This globally recognised open source standard is not covered by the US export restrictions.) Alibaba's chip-making subsidiary, Pingtoug, has launched its first product, an open source chip processor dubbed XuanTie 910.



The processor uses RISC-V, an open source hardware instruction set architecture (ISA) – the key part of the programming infrastructure that decides how a device functions. It can be used in designing chips for 5G telecommunications, artificial intelligence as well as autonomous driving.

The processor could potentially double chip performance while reducing costs of related chip production by more than 50 per cent, Alibaba said.

The processor will soon be available for commercial sale, the company told Chinese daily Caixin, without providing a timetable or price range. To promote innovation, Pingtoug will soon allow global developers to download some of XuanTie 910's code for free on GitHub.

marked as a potential cause of the disruption of biological processes in diseases including cancer.

The third and final project is called 'Pathway-induced multiple kernel learning' or PIMKL. This algorithm uses data sets describing what we currently know when it comes to molecular interactions, in order to predict the progression of cancer and potential relapses in patients.

The source code is now available on the projects' websites.

China's very own open source software foundation coming up soon

The ongoing US-China trade war is actually making Chinese companies stronger, by reducing their technological dependence on the US. After Huawei launched its own operating system, HarmonyOS at the recent Huawei Developer Conference, it plans to partner with several companies to set up China's first open source software foundation to grow the nation's open source software community, *China Daily* reported.

According to the report, the foundation may be established within one or two months.

The foundation plan came after GitHub, the world's largest host of source code, in July prevented users in Iran and other sanctioned nations from accessing portions of its services.

"If China does not have its own open source community to maintain, manage and host the open source code, the domestic software industry will be very vulnerable to uncontrollable factors," Wang Chenglu, president of the software department at Huawei's consumer business group, was quoted as saying.

According to Wang, the first open source software foundation in China will be non-profit, completely public and open to all companies and software developers.

Spain set to create world's first free and open source IoT network

Fossa Systems, a non-profit youth association based out of Madrid, Spain, is developing the country's first open source pico satellite (below 1.6kg), which it plans to launch into orbit in October 2019.

Measuring just 5cm³ and weighing only 250g, it will be one of the smallest satellites ever launched into space and will create the world's first free and open source IoT network, the association wrote on its official website.

Dubbed FossaSat-1, the satellite will provide free IoT connectivity by communicating LoRa RTTY signals through low-power RF based LoRa modules.

The project's plans include testing the new LoRa spread spectrum modulation



using less than US\$ 5 worth of receivers, with the goal of democratising telecommunications worldwide. According to sources at Fossa Systems, the satellite's development and launch will cost less than US\$ 33,313.

"Our objective is to democratise access to space, both regarding telecommunications and access to in-

orbit hardware, by using pico satellites that can fit in your pocket and be launched for less than the cost of a car," the association said.

"FossaSat-1 will provide free and open source IoT communications for the globe using inexpensive LoRa modules, with which students will be able to communicate with a satellite for under US\$ 22," it added.

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